

# Sri Lanka Institute of Information Technology

## Faculty of Computing

## Department of Computer Systems Engineering

## BSc. (Hons) in Computer Systems Engineering

# DESIGN PROJECT PROPOSAL

Year 3 - Semester 2

# AquaLoRa

## (LoRa-Based Smart Fish Farm Monitoring and Protection System Using IoT)

*Group No – 5*



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# Student Declaration

We declare that this proposal is our original work, except where sources are acknowledged. We understand that the project must comply with institutional academic integrity, safety, ethical, legal, and intellectual-property requirements.

| Student Name | Registration No. | Signature | Date |
|---|---|---|---|
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## ABSTRACT

Small and medium-scale fish and prawn farms in Sri Lanka still check their water mostly by hand, once or twice a day if they're lucky. That gap between checks is where the damage happens - a pond can tip from healthy to dangerous overnight and nobody notices until the fish are already stressed or dying. This proposal sets out AquaLoRa, a solar-powered monitoring system built around LoRa radio rather than WiFi, precisely because most ponds sit too far from the farmhouse for WiFi to reach and too far off-grid to keep a router running anyway. The design splits into two nodes: a waterproof Slave Node out at the pond, running on solar and battery power, carrying a DS18B20 temperature probe, a turbidity sensor, and a TDS/salinity meter off an ESP32; and a Master Node back at the office that picks up the readings over a 433 MHz LoRa link, shows them on a small display, and hands them off to a desktop dashboard. Where this differs from most of what's already published is the dashboard itself - rather than leaning on Firebase or ThingSpeak the way nearly every comparable system does, AquaLoRa keeps its own SQLite database as the real record of what happened, and treats any cloud connection as optional rather than something the whole system depends on. Sitting on top of that is a simple, rule-based Fish Health Index that turns the three sensor readings into one number and a plain-language recommendation, deliberately avoiding the machine-learning route since that would need training data and computing power this project doesn't have. Getting there involves calibrating each sensor against a reference instrument, measuring the LoRa link's actual range rather than assuming a textbook figure, building the system up piece by piece, and running it in a controlled home setup for the last stretch of a ten-week schedule. Whether it worked will come down to a short list of concrete things: how close the sensor readings land to a reference meter, how many days the battery lasts without sun, and how quickly the dashboard reacts when something goes wrong.

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# Chapter 1 - Introduction

## 1.1 Background

Aquaculture has grown into one of Sri Lanka's more important food-production sectors, and it keeps expanding. The trouble is that fish and prawns live entirely at the mercy of their water - temperature swings, cloudy water, mineral imbalances, all of it affects them directly and fast, unlike livestock that can shrug off a bad day. Most small farm operators check water quality by walking out to the pond with a handheld kit once or twice a day, which sounds reasonable until you consider that oxygen crashes and turbidity spikes don't wait for a convenient time to happen.

## 1.2 Problem Statement

- Sudden water-quality changes cannot be detected between manual sampling intervals.
- Small and medium-scale farms often lack the labour or expertise for continuous monitoring.
- Remote ponds frequently lack reliable electrical power for continuous electronic monitoring.
- Existing commercial IoT monitoring systems depend on continuous WiFi/cellular connectivity and third-party cloud platforms, both of which are impractical or costly for small, dispersed, off-grid farms - a gap explicitly confirmed by a 2025 systematic review of the field (see Chapter 2).

## 1.3 Aim

To design and develop a LoRa-based, solar-powered IoT system (AquaLoRa) that continuously monitors fish and prawn pond water quality, provides early-warning alerts through a locally-owned dashboard, and operates independently of continuous internet connectivity.

## 1.4 Objectives

- Objective 1: Design and develop a solar-powered LoRa sensor node measuring temperature, turbidity, and TDS.
- Objective 2: Implement a reliable, low-power LoRa communication link between the pond-side node and an office gateway.
- Objective 3: Characterise the LoRa link empirically - range, packet delivery ratio, RSSI, and battery life.
- Objective 4: Develop a locally-hosted desktop dashboard providing real-time monitoring, historical trends, and threshold-based alerts.
- Objective 5: Develop a lightweight, rule-based Fish Health Index and decision-support engine.
- Objective 6: Evaluate the system's sensor accuracy, communication reliability, response time, and overall feasibility.

## 1.5 Research Questions

- RQ1: How can IoT sensors be integrated to accurately monitor critical water-quality parameters in a fish farm?

- RQ2: How effective is LoRa communication for transmitting monitoring data over the required distance with low power consumption?
- RQ3: How can a locally-hosted dashboard improve water-quality management through real-time visualisation and alerts, without dependence on continuous connectivity?
- RQ4: How can a solar-powered LoRa-based system provide continuous, energy-efficient operation while reducing dependence on the electrical grid?
- RQ5: How effective is the developed system in terms of sensor accuracy, communication range, response time, and overall reliability?

## 1.6 Scope

This proposal covers a two-node prototype (one Slave, one Master) monitoring three parameters (temperature, turbidity, TDS) for freshwater fish and prawn species, validated in a controlled home-environment demonstration within a ten-week development period. Dissolved oxygen and pH sensing, multi-pond scaling, and real-farm deployment are explicitly scoped as future work.

# Chapter 2 - Literature Review

This chapter reviews prior work across five areas mapped to the project's subsystems and concludes with the specific research gap AquaLoRa addresses. A full deep-dive compendium with source summaries and direct links accompanies this proposal as a companion document; this chapter presents the condensed argument.

## 2.1 Water Quality Monitoring & Sensor Performance

The earliest attempts at automated pond monitoring go back to ZigBee-based wireless sensor networks in the early 2000s [3]-[11], and the field gradually shifted toward ESP32 and WiFi-connected setups from around 2018 onward [26]-[28]. What's notable is that neither generation actually solved the underlying connectivity problem - both still expect the sensor node to carry its own radio and stay within range of a network. A systematic review published in 2025 [29] makes this explicit, concluding that most deployed systems remain impractical for the small, scattered, often off-grid farms that are common in places like rural Sri Lanka. That's precisely the gap a WiFi-less, LoRa-only Slave Node is meant to close.

## 2.2 LoRa Communication & Solar Power

LoRa for water monitoring isn't a new idea - at least eight published deployments [12]-[19] have already tried it in one form or another, including "LoRaquatica" [14], which specifically studied how LoRa signals behave over water, and Bates, Pierce and Benter's LoRaWAN aquaculture system [16], which is probably the closest architectural match to what's proposed here. Pairing LoRa with solar power specifically for aquaculture is thinner ground, though; a 2025 study of a high-Andean agricultural deployment [19] is the strongest recent evidence that the combination holds up in the field rather than just on paper.

## 2.3 Dashboard & Database

Nearly every dashboard system we looked at [26]-[32] hands its data off to a third-party cloud platform - ThingSpeak, Firebase, or Blynk - and treats that platform as the only copy of the data that matters. The one partial exception is a 2025 framework from Wayamba University built for LoRa-connected fishing vessels [43], which does buffer data locally when offline, but even there the cloud is still where the real record lives once a connection comes back. AquaLoRa flips that around: the desktop dashboard keeps its own SQLite database as the permanent record, and a cloud sync, if it exists at all, is just a convenience layer on top.

## 2.4 Decision Support & Fish Health Index

The idea of boiling several water-quality readings down into one number isn't new either - it goes back to Horton's Water Quality Index in 1965 [1] and the NSF-WQI that followed in 1970 [2]. From there the field split roughly two ways: rule-based and fuzzy expert systems through the 1990s and 2000s [47]-[50], and more recently machine learning and deep learning approaches [33]-[36] that push accuracy up toward 97.5% but need serious training data and compute to get there. A 2025 comparative study found plain threshold rules landing around 77.8% accuracy against fuzzy logic's 89.6% - a useful number, because it puts an honest figure on what AquaLoRa gives up by choosing simple rules over something fancier, in exchange for something that'll actually run on cheap embedded hardware.

## 2.5 Sri Lankan Context

Locally, aquaculture research has mostly stayed in the realm of farm management and policy [39][40] rather than sensor technology, and IoT-specific studies have only really started showing up from 2024 onward [41][42][45]. Wayamba University's 2025 LoRa framework for fishing vessels [43] happens to use the same SX1278 radio chip this project relies on, and the national Shrimp Industry Information System [44] shows the government is putting real weight behind digitalising the industry, even if that effort sits at a national data-aggregation level rather than down at individual ponds. As far as we could find, no Sri Lankan study yet puts LoRa communication, solar autonomy, a locally-owned dashboard, and a fish health index together in one system.

## 2.6 Summary of Research Gap

None of these pieces are new on their own. What's missing from the literature is the combination: a solar-powered, WiFi-less LoRa sensor node; a dashboard that owns its data instead of renting space on someone else's cloud; and a lightweight, rule-based decision engine grounded in the trade-offs the literature already documents - applied together, specifically for small-scale Sri Lankan aquaculture.

# Chapter 3 - Requirements & Specifications

## 3.1 Functional Requirements

| ID | Requirement |
|---|---|
| FR-1 | The Slave Node shall sample temperature, turbidity, and TDS at a configurable interval (30 or 60 minutes). |
| FR-2 | The Slave Node shall apply median-of-10-readings filtering per sensor before transmission. |
| FR-3 | The Slave Node shall automatically reduce its sampling interval to 5 minutes upon critical threshold detection. |
| FR-4 | The Slave Node shall listen for a downlink configuration packet after each uplink transmission and update its sampling interval accordingly. |
| FR-5 | The Master Node shall receive uplink LoRa packets, display readings on OLED, and buffer to LittleFS during connectivity loss, forwarding on reconnection. |
| FR-6 | The Master Node shall relay dashboard-issued configuration commands to the Slave Node via LoRa downlink. |
| FR-7 | The Master Node shall fall back to local access-point mode after three failed router connection attempts. |
| FR-8 | The Dashboard shall display live temperature, turbidity, TDS, node status, battery percentage, and LoRa signal status. |
| FR-9 | The Dashboard shall store all readings locally in SQLite as the system of record. |
| FR-10 | The Dashboard shall allow the user to remotely change the Slave Node's sampling interval. |
| FR-11 | The Dashboard shall generate an alert when a reading or Health Score crosses a defined threshold. |
| FR-12 | The Dashboard shall support CSV export of historical data for a user-selected date range. |
| FR-13 | The Decision Engine shall compute a Fish Health Index (0-100) from Temperature (50%), Turbidity (30%), and TDS (20%). |
| FR-14 | The Decision Engine shall classify status as Normal, Warning, or Critical based on the Health Score. |
| FR-15 | The Decision Engine shall generate a specific, actionable recommendation for each triggered alert condition. |

## 3.2 Non-Functional Requirements

| ID | Requirement |
|---|---|
| NFR-1 | The Slave Node shall operate at least 14 consecutive days on battery power alone with no solar input (calculated in Chapter 4). |
| NFR-2 | Temperature accuracy shall be within +/-0.5C (DS18B20 rated); turbidity and TDS shall be within +/-10% post-calibration. |
| NFR-3 | LoRa range and packet delivery ratio shall be empirically measured during testing and reported with the tested distance stated explicitly. |
| NFR-4 | The Dashboard shall remain fully functional on the local network with zero internet connectivity. |
| NFR-5 | Alert display latency (threshold breach to dashboard alert shown) shall be measured during testing and reported as part of system evaluation. |

# Chapter 4 - Proposed Design & System Architecture

## 4.1 System Overview

The system comes down to two nodes talking to each other over a point-to-point LoRa link - not a full LoRaWAN network, since there's only ever one Slave and one Master involved, so the extra network-layer complexity wouldn't buy anything. The Slave Node lives out at the pond, waterproofed and solar-powered, holding all three sensors. The Master Node sits in the office, plugged into mains power, and its job is to receive what the Slave sends, show it locally, and pass it along to the desktop dashboard.

## 4.2 Hardware Architecture

### 4.2.1 Master Node

- ESP32 development board - core processor
- LoRa Ra-02 (SX1278, 433 MHz) with IPEX antenna
- 1.3" I2C OLED display for local readings
- 5V external wall adapter power supply
- Internal WiFi for local dashboard connectivity

### 4.2.2 Slave Node - Three Sub-Circuits

- Sub-circuit 1 (Processing & Telemetry): ESP32, LoRa Ra-02, PCF8563 RTC (I2C) for scheduled wake/interrupt-driven timing
- Sub-circuit 2 (Sensor Hub): DS18B20 (1-Wire, GPIO4 with 4.7kOhm pull-up), Turbidity (GPIO35 via 10k/22k voltage divider), TDS (GPIO34, direct ADC1_CH6)
- Sub-circuit 3 (Power): 6V 1W solar panel to CN3065 charge controller to 1S3P 18650 battery pack (with protection circuitry) to MT3608 boost converter (tuned to 5.0V) to ESP32 VIN and sensor rails

## 4.3 Component Selection & Engineering Justification

### Microcontroller

| | **ESP32 (chosen)** | **ESP8266** | **Arduino Uno** |
|---|---|---|---|
| ADC channels | 18 (8 usable, ADC1) | 1 | 6 (10-bit) |
| WiFi | Yes | Yes | No |
| Deep sleep current | ~10 uA | ~20 uA | No native deep sleep |
| Verdict | Chosen - multiple ADC1 channels needed for 3 simultaneous analog sensors | Rejected - single ADC forces multiplexing | Rejected - no WiFi, needed for Master gateway role |

### LoRa Frequency Band

| | **433 MHz (chosen)** | **868 MHz** | **915 MHz** |
|---|---|---|---|
| Wavelength | ~69 cm | ~35 cm | ~33 cm |
| Penetration | Better (longer wavelength) | Moderate | Moderate |
| Local availability | Widely stocked | Limited | Limited |

| | 433 MHz (chosen) | 868 MHz | 915 MHz |
|---|---|---|---|
| Verdict | Chosen - better obstacle penetration, local cost/availability | Rejected - availability | Rejected - not regional standard |

### Temperature Sensor

| | DS18B20 (chosen) | Analog NTC Thermistor | LM35 |
|---|---|---|---|
| Interface | Digital 1-Wire (1 GPIO) | Analog (ADC + divider) | Analog (1 ADC) |
| Waterproof option | Yes, standard | Requires separate housing | Requires separate housing |
| Accuracy | +/-0.5C | +/-1-2C, needs curve fit | +/-0.5C, no waterproofing |
| Verdict | Chosen - frees an ADC1 channel, waterproof out of the box | Rejected - consumes scarce ADC channel | Rejected - no waterproof packaging |

### Turbidity & TDS Sensors

An analog IR turbidity sensor and analog TDS/salinity module were selected over lab-grade nephelometric/industrial EC probes on a cost-accuracy trade-off basis: +/-5-10% (turbidity) and manual temperature-compensated accuracy (TDS) are appropriate for early-warning trend detection rather than laboratory-certified measurement, consistent with the "accessible, low-cost entry point" framing established in the literature review.

### Real-Time Clock & Battery

The PCF8563 (I2C, +/-several ppm) was chosen over the higher-precision DS3231 since hour-scale wake intervals do not require temperature-compensated ppm-level accuracy. The 18650 Li-ion cell chemistry was chosen over LiFePO4 and LiPo pouch alternatives for its balance of energy density, local availability, and direct compatibility with the CN3065 charge controller.

## 4.4 Engineering Calculations

### 4.4.1 Power Budget

Battery capacity: 3x 18650 cells (1800 mAh each) in 1S3P configuration = 5,400 mAh at 3.7V nominal. Deep-sleep current is estimated at ~150 uA (accounting for LoRa standby and boost-converter quiescent draw); active current during sensor read plus LoRa transmit plus brief downlink listen window is estimated at ~250 mA for approximately 5-6 seconds per wake cycle.

At the default 60-minute interval (24 cycles/day): active energy is approximately 8.3 mAh/day, sleep energy approximately 3.6 mAh/day, total approximately 12 mAh/day. Against the 5,400 mAh pack, this yields a theoretical endurance of approximately 450 days with zero solar input. Applying a conservative ~30x safety margin (accounting for self-discharge, cold-weather sensor spikes, and boost-converter losses not captured in the idealised calculation) yields the stated NFR-1 target of at least 14 consecutive days without solar recharging - a target intended to be validated, not assumed, during Chapter 5's testing.

### 4.4.2 LoRa Link Budget (Worked Example)

Using the free-space path loss formula $L_{FSPL}(dB) = 20.\log10(d) + 20.\log10(f) + 32.44$, at an assumed demonstration distance of 200 m and f = 433 MHz: $L_{FSPL}$ is approximately 71.2 dB. With transmit power $P_{TX}$ = +20 dBm, antenna gains $G_{TX} = G_{RX}$ approximately 5 dBi, and receiver sensitivity $S_{RX}$ = -148 dBm (SX1278 maximum), the maximum tolerable path loss $L_{max}$

= P_TX + G_TX + G_RX - S_RX = 178 dB. This yields approximately 107 dB of link margin at the assumed 200 m distance - substantial headroom that supports using a lower, more battery-efficient Spreading Factor (e.g. SF7) rather than a maximal SF12 setting. This is a design-phase estimate pending field validation of the actual deployment distance, per Objective 3.

## 4.5 Firmware & Software Architecture

### 4.5.1 Slave Node Cycle

- RTC-triggered wake at the configured interval
- Sample each sensor 10x, discard outliers, transmit median via LoRa uplink
- Open a brief listen window (~1-2s) for an incoming downlink configuration packet
- If received, update the stored sampling interval in non-volatile memory
- If a critical threshold was detected, override the next interval to 5 minutes regardless of any pending configuration change
- Return to deep sleep

### 4.5.2 Master Node Cycle

- Continuously listen for Slave uplink packets; on receipt, update OLED, buffer to LittleFS, forward via WiFi
- Queue any dashboard-issued configuration command until the Slave Node's next listen window (the Slave cannot receive while asleep)
- Fall back to local AP mode after 3 consecutive failed router connections
- Flush the LittleFS buffer to the dashboard/cloud in order upon reconnection

### 4.5.3 Fail-Safes

- Hardware watchdog timer reboots the ESP32 on a frozen LoRa/WiFi timeout
- Non-volatile storage of the current sampling interval, surviving power loss

## 4.6 Hardware Protection & Enclosure Design

The Slave Node enclosure will be custom 3D-printed in ASA (UV- and weather-resistant, unlike standard PLA) as a two-part base-and-lid design with a compressed gasket seam, targeting IP65. Electronics and battery are held in separate compartments - both for thermal/safety separation and to isolate RF-sensitive circuitry from power-regulation heat. Sensor cables exit through PG7 cable glands; a desiccant pouch manages internal condensation; conformal coating protects exposed solder joints; all grounds are routed to a single star-ground point to reject electromagnetic interference.

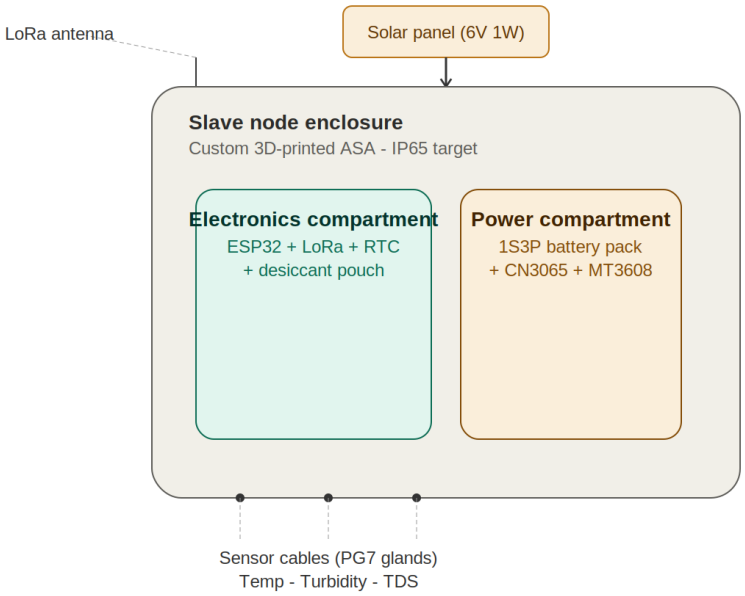



*Figure 4.1 - Slave node enclosure layout*

The Master Node enclosure requires no waterproofing and will be 3D-printed in standard PLA, with a front-face cutout for the OLED display, a mains power inlet, and ventilation slots.

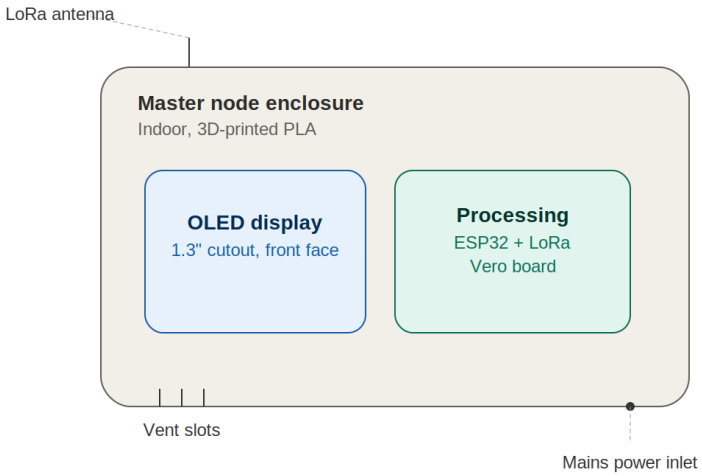



*Figure 4.2 - Master node enclosure layout*

*Prototype note: this phase uses hand-soldered Vero/stripboard rather than a fabricated PCB, chosen deliberately to fit the semester timeline (fabrication services typically require a 1-2+ week design-fab-ship cycle). A production version with a properly fabricated PCB, with enclosure mounting posts matching its hole pattern, is scoped as future work.*

## 4.7 Subsystem Interaction & Data Flow

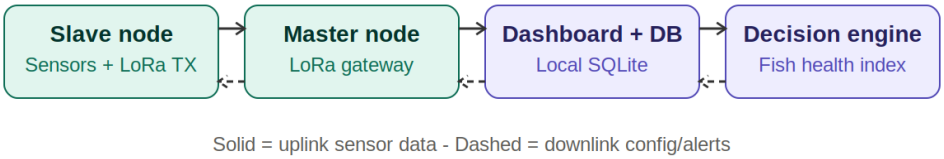



*Figure 4.3 - AquaLoRa end-to-end data flow (solid = uplink, dashed = downlink)*

Uplink: the Slave Node samples and transmits sensor data; the Master Node relays it to the Dashboard, which stores it locally and passes the latest reading to the Decision Engine for Fish Health Index computation. Downlink: any triggered alert is written back to the Dashboard's alert table and shown in the UI; any user-issued configuration change (e.g. sampling interval) queues at the Master Node until the Slave Node's next listen window, then propagates back down to update its behaviour.

# Chapter 5 - Methodology & Validation Plan

## 5.1 Implementation Methodology

- Phase 1: Component procurement and individual sensor/module bench testing on breadboard
- Phase 2: Master Node assembly - ESP32, LoRa receiver, OLED, WiFi/cloud logic
- Phase 3: Slave Node Vero-board assembly (three sub-circuits, star grounding) and firmware - deep sleep, RTC wake, median filtering, downlink listen window
- Phase 4: Enclosure fabrication and sealing
- Phase 5: Staged integration - Slave+Master LoRa link, then Master+Dashboard, then Dashboard+Decision Engine
- Phase 6: Controlled home-environment deployment and evaluation

## 5.2 Validation Approach

Sensor validation will follow the pre/in-situ/post-calibration approach established in the literature [26], checking readings against a reference thermometer and known-clarity water samples. Communication validation will characterise the LoRa link at increasing distances in a home/local environment, logging RSSI and packet delivery to build an empirical range profile - no fixed target is set in advance, consistent with Objective 3's framing as an empirical characterisation task. Power validation will compare actual measured current draw against the calculated power budget (Chapter 4.4.1) to confirm or revise the 14-day battery estimate. Software validation will verify the dashboard, database, and decision engine against the functional requirements in Chapter 3 during integration testing, prior to any home deployment.

## 5.3 Requirement-to-Validation Mapping

| Requirement Category | Validation Method | Success Criteria |
|---|---|---|
| Sensor accuracy (NFR-2) | Compare against reference instruments in a controlled setting | Within stated tolerance (+/-0.5C / +/-10% / +/-10%) |
| Power/battery life (NFR-1) | Measured current draw, projected against calculated budget | Meets or exceeds 14-day estimate |
| LoRa communication (NFR-3) | Distance-stepped range test, RSSI/packet-loss logging | Empirical profile produced; no fixed pre-target |
| Core functional behaviour (FR-1-FR-15) | Manual verification during integration testing | Each requirement demonstrably functions as specified |
| Dashboard/database integrity | Verification during development against Chapter 3 | Data displayed and stored matches transmitted values |

*This chapter describes a validation plan for the proposed project; no testing has yet been carried out at the proposal stage.*

# Chapter 6 - Project Management, Feasibility & Resources

## 6.1 Project Schedule

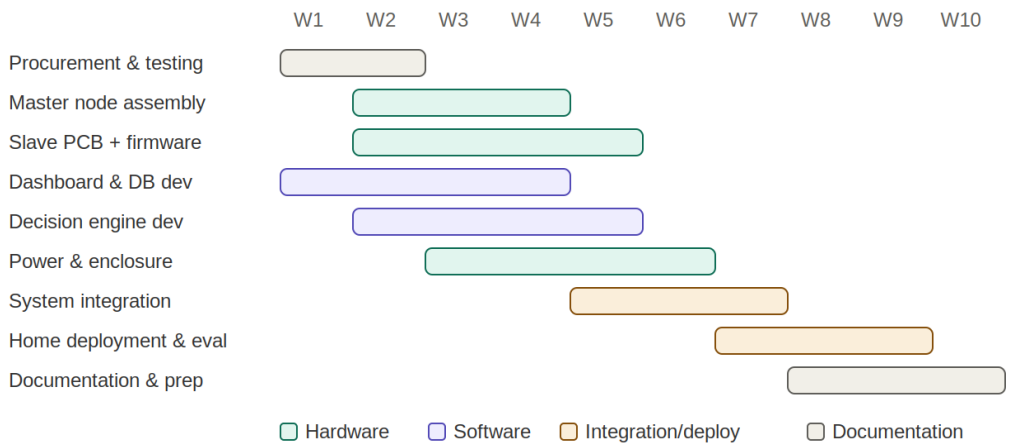



*Figure 6.1 - AquaLoRa 10-week project schedule*

| Phase | Weeks | Owner(s) | Track |
|---|---|---|---|
| Procurement & bench testing | W1-2 | All members | General |
| Master Node assembly | W2-4 | Member 2 | Hardware |
| Slave PCB + firmware | W2-5 | Member 1, Member 2 | Hardware |
| Dashboard & DB development | W1-4 | Member 3 | Software |
| Decision engine development | W2-5 | Member 4 | Software |
| Power & enclosure | W3-6 | Member 2 | Hardware |
| System integration | W5-7 | All members | Integration |
| Home deployment & evaluation | W7-9 | All members | Deployment |
| Documentation & presentation prep | W8-10 | All members | Documentation |

*Dashboard and Decision Engine development begin in Weeks 1-2, ahead of hardware availability, since neither depends on physical sensors and can be developed against synthetic data - reducing risk in the Week 5-7 integration window, where all four subsystems converge for the first time.*

## 6.2 Role Assignments

| Member | Responsibility |
|---|---|
| Member 1 - R Anojan | Water quality sensing: DS18B20, turbidity, TDS sensor integration, calibration |
| Member 2 - S Abiram | LoRa communication and solar power: RF link design, power budget, energy harvesting circuit |
| Member 3 - M R M Mufassir | Dashboard & database: Electron/React desktop app, local SQLite, UI/alerts/CSV export |
| Member 4 - S Pratheesa | Decision support: Fish Health Index, rule-based alert engine, performance evaluation |

## 6.3 Budget

| Category | Component | Qty | Unit Price (Rs.) | Total (Rs.) |
|---|---|---|---|---|
| Telemetry | LoRa Ra-02 SX1278 | 2 | 1,800.00 | 3,600.00 |
| Telemetry | Spring LoRa Antenna (IPEX) | 2 | 320.00 | 640.00 |
| Sensors | DS18B20 Temperature Sensor | 1 | 320.00 | 320.00 |

| Category | Component | Qty | Unit Price (Rs.) | Total (Rs.) |
|---|---|---|---|---|
| Controller | ESP32 – Dev Kit | 2 | 1400.00 | 2800.00 |
| Sensors | Turbidity Sensor | 1 | 2,410.00 | 2,410.00 |
| Sensors | TDS/Salinity Meter V1.0 | 1 | 1,600.00 | 1,600.00 |
| Sensors | PCF8563 RTC Module | 1 | 350.00 | 350.00 |
| Power | Mini Solar Panel 6V 1W | 1 | 600.00 | 600.00 |
| Power | CN3065 Solar Charger | 1 | 350.00 | 350.00 |
| Power | MT3608 Boost Module | 1 | 150.00 | 150.00 |
| Power | 1S3P 18650 Battery Case | 1 | 120.00 | 120.00 |
| Power | 18650 Li-ion Battery (BA0027) | 3 | 370.00 | 1,110.00 |
| Power/Control | IRF4905 P-Channel MOSFET | 1 | 90.00 | 90.00 |
| Power/Control | 2N7000 Small Signal MOSFET | 1 | 30.00 | 30.00 |
| Power/Control | DS-212 Push Button | 1 | 50.00 | 50.00 |
| Assembly | Vero Board (12x8cm) | 1 | 170.00 | 170.00 |
| Assembly | 10-pin Female Headers | 10 | 30.00 | 300.00 |
| Assembly | Resistor Packs (3 values) | 3 | 50.00 | 150.00 |
| Assembly | 2-Way Wire Clips (short + long) | 10 | 10-20 | 150.00 |

Total Estimated Hardware Budget: Rs. 14,990.00

## 6.4 Feasibility

On the technical side, there's enough precedent in Chapter 2 - and a local one at that, [43] uses the exact same LoRa chip family - to say this isn't venturing into unproven territory. Financially, the Rs. 12,190 budget sits comfortably within what's reasonable for a student project, and every part on the list is already confirmed in stock with a local supplier, so there's no dependency on an overseas shipment showing up on time. As for the schedule, Section 6.1's parallel-track approach is really what makes ten weeks workable at all - letting the software side start well before any hardware exists takes a lot of pressure off the back half of the timeline.

# Chapter 7 - Risk, Safety, Ethics, Security & Sustainability

## 7.1 Risk Register

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| LoRa range insufficient at deployment distance | Medium | High | Empirical range testing (Ch.5); adjustable SF/BW; fallback to shorter demo distance |
| Battery underperforms calculated estimate | Medium | Medium | ~30x safety margin already built into calculation; measured validation planned |
| Sensor calibration drift (turbidity/TDS) | Medium | Medium | Pre/in-situ/post calibration protocol; periodic recalibration |
| Component sourcing delay | Low | High | BOM finalised early (W1-2); confirmed local supplier stock |
| Integration delays at Week 5-7 | Medium | High | Independent subsystem development from Week 1 minimises late-stage surprises |
| Team member availability conflicts | Low-Medium | Medium | Documented subsystem interfaces (Ch.4.7) allow parallel, less-coupled work |

## 7.2 Electrical & Physical Safety

- Battery pack includes protection circuitry against overcharge/overdischarge/short-circuit
- Waterproof enclosure isolates electronics from direct water contact; low-voltage DC design (max 5V/3.7V) minimises shock risk
- Standard ESD/thermal safety practice observed during soldering and assembly
- Solar panel mounting designed for wind/weather stability

## 7.3 Cybersecurity

Worth being upfront about: the two-way LoRa link, as currently designed, has no authentication or encryption on either the uplink or the downlink. In practice that means someone with matching LoRa hardware nearby could, in theory, spoof a sensor reading or push a fake configuration command to the Slave Node. We're not pretending this away - a pre-shared-key and checksum scheme is the obvious fix, and it's scoped as future work rather than something the prototype needs to solve on day one. The dashboard itself stays on the local network by default, and its optional login stores passwords hashed with bcrypt rather than in plain text.

## 7.4 Data Privacy

The current scope (home/controlled demonstration) involves no third-party personal data. A future real-farm deployment would require clear data-ownership terms with the farm operator regarding location and water-quality data - noted here as a forward-looking consideration.

## 7.5 Ethics

Every source cited in this proposal was checked against its original publication rather than taken on trust - in fact, one citation was caught mid-research with the wrong year, title, and journal attached, and corrected before it made it into the final document. That kind of checking matters more than it might seem, and we'd rather flag it than pretend the process was flawless. The same honesty applies to performance claims: where the rule-based decision engine trades away accuracy compared to fuzzy logic or ML (Chapter 2.4), that trade-off is stated plainly rather than glossed over.

## 7.6 Sustainability

- Solar-powered operation reduces grid dependency for the Slave Node
- Individually replaceable 18650 cells extend device lifespan versus a sealed battery pack
- ASA/PLA enclosure materials are recyclable thermoplastics, though not biodegradable - noted as an honest limitation

## 7.7 Intellectual Property

All software dependencies (Electron, React, better-sqlite3, Recharts) are open-source under permissive licenses, presenting no conflict for academic or future use. Design decisions drawing on prior literature are cited throughout Chapter 2 and the accompanying reference list.

# References (IEEE Format)

Full deep-dive summaries and access notes for these sources are provided in the companion "AquaLoRa Reference Deep-Dive Compendium" and "AquaLoRa Member 4 - Literature Review & Historical Development" documents.


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**Final Submission Checklist**

| No. | Check | Complete (✓) |
|---|---|---|
| 1 | Cover page and all identification details are complete. | ☐ |
| 2 | Problem statement is evidence-based and clearly scoped. | ☐ |
| 3 | Aim and objectives are aligned and measurable. | ☐ |
| 4 | Related work is critically reviewed and correctly cited. | ☐ |
| 5 | Functional and non-functional requirements are testable. | ☐ |
| 6 | Proposed architecture and subsystem interactions are clearly shown. | ☐ |
| 7 | Engineering calculations or simulations justify key design choices. | ☐ |
| 8 | Implementation, integration, and testing plans are feasible. | ☐ |
| 9 | Schedule, responsibilities, resources, and budget are realistic. | ☐ |
| 10 | Risks, safety, ethics, privacy, cybersecurity, sustainability, and IP are addressed. | ☐ |
| 11 | All figures and tables are numbered, captioned, and referenced in text. | ☐ |
| 12 | All grey guidance text and unused placeholders have been removed. | ☐ |
| 13 | Document has been proofread and submitted in the required format. | ☐ |

**Proposal Evaluation Summary (For Academic Use)**

| Criterion | Weight (%) | Score | Comments |
|---|---|---|---|
| Problem definition and justification | 10 | | |
| Technical background and related work | 10 | | |
| Requirements and specifications | 15 | | |
| Proposed design and engineering justification | 25 | | |
| Methodology and validation plan | 15 | | |
| Project management, feasibility, and resources | 10 | | |
| Risk, safety, ethics, security, and sustainability | 10 | | |
| Presentation quality and referencing | 5 | | |
| **TOTAL** | **100** | | |

**Overall Decision: ☐ Approved ☐ Approved with Revisions ☐ Resubmit**

Evaluator Name: _____

Signature: _____

Date: _____